

# 第 12 章 微分方程

**例1** 已知 $y = y(x)$ 在 $x$ 处增量 $\Delta y = \frac{y\Delta x}{1+x^2} + \alpha$ , 当 $\Delta x \rightarrow 0$ 时,  $\alpha$ 为 $\Delta x$ 的高阶无穷小,  $y(0) = \pi$ , 求 $y(1)$ .

**解** 原式可化为:

$$\frac{\Delta y}{\Delta x} = \frac{y}{1+x^2} + \frac{\alpha}{\Delta x}, \text{ 两边取极限 } \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \left( \frac{y}{1+x^2} + \frac{\alpha}{\Delta x} \right)$$

建立微分方程为:

$$\frac{dy}{dx} = \frac{y}{1+x^2}, \text{ 分离变量得 } \frac{dy}{y} = \frac{1}{1+x^2} dx, \text{ 解得}$$

$$\ln y = \arctan x + C, \text{ 代入条件 } y(0) = \pi \text{ 得 } C = \ln \pi.$$

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$$\ln y = \arctan x + C, \text{ 代入条件 } y(0) = \pi \text{ 得 } C = \ln \pi.$$

$$\text{从而 } \ln y(1) = \arctan 1 + \ln \pi = \frac{\pi}{4} + \ln \pi, \text{ 于是有}$$

$$y(1) = e^{\frac{\pi}{4} + \ln \pi} = \pi e^{\frac{\pi}{4}}.$$

**例2** 设 $f(x)$ 在 $x > 0$ 可微, 且 $f(x) = 1 + \frac{1}{x} \int_1^x f(t) dt$ , 求 $f(x)$ .

解: 原方程可化为:

$$xf(x) = x + \int_1^x f(t) dt, \text{ 两边求导可得 } f'(x) = \frac{1}{x}$$

可解得 $f(x) = \ln x + C$ , 注意到隐藏条件 $f(1) = 1$ , 代入可得 $C = 1$ , 故 $f(x) = \ln x + 1$ .

例3 设 $f(u)$ 在 $(0, +\infty)$ 内具有二阶导数, 且 $z = f(\sqrt{x^2 + y^2})$

满足 $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$ , 若 $f(1)=0$ ,  $f'(1)=1$ , 求 $f(u)$ 的表达式.

解 
$$\frac{\partial z}{\partial x} = f'(u) \frac{x}{\sqrt{x^2 + y^2}} = \frac{xf'(u)}{u}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{f'}{u} + \frac{x^2}{u^3} (uf'' - f')$$

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \frac{f' + uf''}{u}$$

以题意有  $f' + uf'' = 0$  解得通解为  $f(u) = C_1 \ln u + C_2$

代入初始条件得  $C_1 = 1, C_2 = 0$  所以  $f(u) = \ln u$

例4 已知 $u(x, y, z) = u(\sqrt{x^2 + y^2 + z^2})$ 有连续的

二阶导数, 且满足 $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$ , 求 $u(x, y, z)$

解 
$$\frac{\partial u}{\partial x} = u'(r) \frac{2x}{2\sqrt{x^2 + y^2 + z^2}} = \frac{xu'(r)}{r}$$

$$\frac{\partial u}{\partial y} = \frac{yu'(r)}{r} \quad \frac{\partial u}{\partial z} = \frac{zu'(r)}{r}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{u'(r)}{r} + x \frac{u''(r)r - u'}{r^2} \cdot \frac{x}{r} = \frac{u'}{r} + \frac{x^2}{r^3} (ru'' - u')$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{3u'}{r} + \frac{ru'' - u'}{r^3} r^2$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{3u'}{r} + \frac{ru'' - u'}{r^3} r^2 = 0$$

$$ru'' + 2u' = 0$$

$$u' = p \quad \frac{2}{r}p + p' = 0 \quad \frac{dp}{dr} = -\frac{2}{r}p \quad \frac{dp}{p} = -\frac{2}{r}dr$$

$$\ln p = -2\ln r + C_1 \quad p = \frac{C_1}{r^2}, \quad \frac{du}{dr} = \frac{C_1}{r^2} \quad u = -\frac{C_1}{r} + C_2$$

$$\text{即 } u(x, y, z) = \frac{C_1}{\sqrt{x^2 + y^2 + z^2}} + C_2$$

例5 设  $r = \sqrt{x^2 + y^2 + z^2} > 0$ , 函数  $u = f(r)$  具有二阶连续的导数, 且  $\operatorname{div}(\operatorname{grad} u) = \ln r$ , 求  $f(r)$ .

解

$$\operatorname{div}(\operatorname{grad} u) = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$$
$$\frac{\partial u}{\partial x} = f'(r) \frac{x}{\sqrt{x^2 + y^2 + z^2}} = \frac{x f'(r)}{r}$$
$$\frac{\partial^2 u}{\partial x^2} = \frac{f'(r)}{r} + x \frac{f'' r - f'}{r^2} \cdot \frac{x}{r} = \frac{f'}{r} + \frac{x^2}{r^3} (r f'' - f')$$
$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{3f'}{r} + \frac{r f'' - f'}{r^3} r^2 = \frac{2f' + r f''}{r}$$

以题意有  $2f' + r f'' = r \ln r$



$$\text{令 } f' = p,$$

$$p' + \frac{2}{r}p = \ln r \text{ 为一阶线性微分方程,}$$

$$p = e^{\int -\frac{2}{r} dr} (C_1 + \int \ln r \cdot e^{\int \frac{2}{r} dr} dr)$$

$$= \frac{1}{r^2} (C_1 + \int r^2 \ln r dr) = \frac{1}{r^2} (C_1 + \frac{1}{3} r^3 \ln r - \frac{1}{9} r^3)$$

$$f(r) = \int \frac{1}{r^2} (C_1 + \frac{1}{3} r^3 \ln r - \frac{1}{9} r^3) dr$$

$$\text{即 } f(r) = -\frac{C_1}{r} - \frac{1}{18} r^2 + \frac{1}{12} r^4 \ln r - \frac{1}{48} r^4 + C_2$$

例6 进行适当变化,将

$y' = y^2 + 2(\sin x - 1)y + \sin^2 x - 2\sin x - \cos x + 1$   
化为可分离变量方程.

解  $\because y' = y^2 + 2(\sin x - 1)y + (\sin x - 1)^2 - \cos x$

$$y' = (y + \sin x - 1)^2 - \cos x$$

$$\text{令 } y + \sin x - 1 = u \quad \frac{dy}{dx} = \frac{du}{dx} - \cos x,$$

代入原方程得

$$\frac{du}{dx} - \cancel{\cos x} = u^2 - \cancel{\cos x}$$

例7 在半空间 $x>0$ , 向量场

$$\vec{A} = \{xf(x), -xyf(x), -e^{2x}z\} \text{ 的 } \operatorname{div} \vec{A} \equiv 0$$

其中 $f(x)$ 在 $(0, +\infty)$ 内具有连续的一阶导数，  
且  $\lim_{x \rightarrow 0^+} f(x) = 1$ , 求 $f(x)$ .

解  $\because \operatorname{div} \vec{A} \equiv 0$  即,  $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 0$ , 则

$$xf'(x) + f(x) - xf(x) - e^{2x} = 0 \quad x > 0$$

$$f'(x) + \left(\frac{1}{x} - 1\right)f(x) = \frac{1}{x}e^{2x} \quad x > 0$$

——一阶线性非齐次方程

解得  $f(x) = \frac{e^x}{x}(e^x + c)$

将条件  $\lim_{x \rightarrow 0^+} f(x) = 1$  代入得 :  $c = -1$

$\therefore f(x) = \frac{e^x}{x}(e^x - 1)$

例8 设 $g(x)$ 在 $(-\infty, +\infty)$ 具有一阶连续导数

$g(0) = 0$ 且能使 $\int_L [ye^{-x} \cos x - yg(x)]dx + g(x)dy$ ,

在全平面上与路径无关, 试求:  $g(x)$

解  $P(x, y) = ye^{-x} \cos x - yg(x)$      $Q(x, y) = g(x)$

$$\because \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \quad \therefore e^{-x} \cos x - g(x) = g'(x)$$

$$\text{故, } g(x) = e^{\int -dx} (c + \int e^{-x} \cos x e^{\int dx} dx) = e^{-x} (c + \sin x)$$

$$\text{将 } g(0) = 0 \text{ 代入得, } c = 0 \quad \therefore g(x) = e^{-x} \sin x$$

例9 已知积分曲线为 $y = c_1x + c_2x^2$ ，求其相应的微分方程。

**解** 此类问题为求解微分方程的反问题，其求解方法是利用微分法消去常数。由通解的定义可知，其对应的微分方程应为二阶微分方程。将所给通解两端关于  $x$  求导得：

$$y' = c_1 + 2c_2x$$

两端关于  $x$  再求导得： $y'' = 2c_2$

$\because$  所给方程是二阶的， $\therefore$  不能再求导。

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$$c_2 = \frac{1}{2} y'' \quad c_1 = y' - 2c_2x = y' - xy'',$$

代入通解表达式可得:  $y = (y' - xy'')x + \frac{1}{2}y''x^2$

即,  $y''x^2 - 2xy' + 2y = 0$

Ex. 连续函数 $f(x)$ 满足 $f(x) = \int_0^{2x} f\left(\frac{t}{2}\right) dt + \ln 2$ , 则

$$f(x) = \underline{e^{2x} \ln 2}.$$

Ex. 设 $\int_L [f(x) - e^x] \sin y dx - f(x) \cos y dy$ 与路径无关,  $f(x)$ 一阶导数连续且 $f(0) = 0$ , 则

$$f(x) = \underline{\frac{1}{2}(e^x - e^{-x})}$$



## 例10 解下列微分方程

$$1. \quad 3e^x \operatorname{tg} y dx + (1 - e^x) \sec^2 y dy = 0$$

$$\text{解} \quad \frac{-3e^x}{1 - e^x} dx = \frac{\sec^2 y}{\tan y} dy$$

$$3 \ln(e^x - 1) = \ln \tan y + \ln c$$

$$c \tan y = (e^x - 1)^3$$

$$2. \quad x(\ln x - \ln y)dy - ydx = 0$$

解  $\frac{dy}{dx} = \frac{y}{x(-\ln \frac{y}{x})}$  令  $y = ux$   $\frac{dy}{dx} = u + x \frac{du}{dx}$

$$u + x \frac{du}{dx} = -\frac{u}{\ln u}, \quad \frac{1}{x} dx = -\frac{\ln u du}{u(1 + \ln u)}$$

$$\frac{1}{x} dx = -\frac{1}{u} \left(1 - \frac{1}{1 + \ln u}\right) du$$

$$\ln x = -\ln u + \ln(1 + \ln u) + \ln c$$

$$\Rightarrow ux = c(1 + \ln u) \Rightarrow y = c\left(1 + \ln \frac{y}{x}\right)$$

另解  $\frac{dx}{dy} = \frac{x}{y} \ln \frac{x}{y}$

令  $x = uy \Rightarrow u + y \frac{du}{dy} = u \ln u$

$$\frac{1}{y} dy = \frac{du}{u(\ln u - 1)}$$

积分  $\ln y = \ln(\ln u - 1)c$

$$y = (\ln u - 1)c \Rightarrow y = \left(\ln \frac{x}{y} - 1\right)c$$

$$3. \quad e^y dx + (xe^y - 2y)dy = 0$$

解  $\frac{dx}{dy} = -x + \frac{2y}{e^y}$

$$x = e^{\int -dy} (c + \int \frac{2y}{e^y} e^{\int dy} dy) = e^{-y} (c + y^2)$$

另解  $P = e^y \quad Q = xe^y - 2y \quad \frac{\partial P}{\partial y} = e^y = \frac{\partial Q}{\partial x}$

$\therefore$  原方程为全微分方程。

$$\begin{aligned} u(x, y) &= \int_0^x e dx + \int_0^y (xe^y - 2y) dy \\ &= x + xe^y - y^2 - x \end{aligned}$$

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$$\therefore e^y x - y^2 = c$$

#### 4. $(xy + 1)ydx - xdy = 0$

解  $\frac{dy}{dx} = y^2 + \frac{1}{x}y$  - 贝努利方程

$$\frac{1}{y^2} \frac{dy}{dx} = \frac{1}{x} \cdot \frac{1}{y} + 1 \quad z = e^{\int -\frac{1}{x} dx} (c + \int (-1)e^{\int \frac{1}{x} dx} dx)$$

$$\text{即, } \frac{d(-\frac{1}{y})}{dx} = \frac{1}{x} \cdot \frac{1}{y} + 1 \quad = \frac{1}{x} (c + \int (-1)x dx)$$
$$\text{令 } z = \frac{1}{y} \quad = \frac{1}{x} (c - \frac{x^2}{2})$$

$$\frac{dz}{dx} = -\frac{1}{x}z - 1 \quad \therefore \frac{1}{y} = \frac{1}{x} (c - \frac{x^2}{2})$$

另解  $xy^2dx + ydx - xdy = 0$

$$\mu(x, y) = \frac{1}{y^2} \Rightarrow xdx + \frac{ydx - xdy}{y^2} = 0$$

$$d\frac{x^2}{2} + d\frac{x}{y} = 0 \qquad \frac{x^2}{2} + \frac{x}{y} = c$$

# 练习

## 例1

设  $y = f(x)$  是方程的  $y'' - 2y' + 4y = 0$  一个解,

若  $f(x_0) > 0$ , 且  $f'(x_0) = 0$ , 则函数  $f(x)$  在点  $x_0$

(a) 取得极大值      (b) 取得极小值

(c) 某个领域内单调增加      (d) 某个领域内单调减少。

答:  $f'(x_0) = 0$  —  $x = x_0$  为驻点,

又  $f''(x_0) - 2f'(x_0) + 4f(x_0) = 0, f(x_0) > 0,$

$\therefore f''(x_0) = -4f(x_0) < 0$ , 故  $x = x_0$  是  $f(x)$  的极大点。

**例2** (1) 试确定以 $y = \sin 2x$ 为特解的二阶常系数线性齐次微分方程。

(2) 试确定以 $y_1 = e^x, y_2 = xe^x, y_3 = 2\cos 2x, y_4 = \sin 2x$ 为特解的四阶常系数线性齐次微分方程。



**解**(1) *i*)特征根为  $r_1 = 2i \Rightarrow r_2 = -2i$

*ii*)特征方程为  $(r - 2i)(r + 2i) = 0$ ,

$$\text{即 } r^2 + 4 = 0$$

*iii*)所求齐次线性方程:  $y'' + 4y = 0$

(2) *i*)由特解  $y_1 = e^x, y_2 = xe^x$  知  $r_{1,2} = 1$ (二重根)

由特解  $y_3 = 2 \cos 2x, y_4 = \sin 2x$  知,  $r_{3,4} = \pm 2i$

$$\text{ii}) (r - 1)^2 (r - 2i)(r + 2i) = 0$$

$$\text{即 } r^4 - 2r^3 + 5r^2 - 8r + 4 = 0$$

*iii*)所求齐次线性方程:

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$$y^{(4)} - 2y''' + 5y'' - 8y' + 4y = 0$$

**例3** 设  $y'' + p(x)y' = f(x)$  有一特解为  $\frac{1}{x}$ , 对应

的齐次方程有一特解为  $x^2$ , 试求:

(1)  $p(x), f(x)$  的表达式;

(2) 此方程的通解.

**解** (1) 由题设可得:

$$\begin{cases} 2 + p(x)2x = 0, \\ \frac{2}{x^3} + p(x)(-\frac{1}{x^2}) = f(x), \end{cases}$$

解此方程组, 得

$$p(x) = -\frac{1}{x}, \quad f(x) = \frac{3}{x^3}.$$

(2) 原方程为  $y'' - \frac{1}{x}y' = \frac{3}{x^3}.$

显见  $y_1 = 1, y_2 = x^2$  是原方程对应的齐次方程的两个线性无关的特解，

又  $y^* = \frac{1}{x}$  是原方程的一个特解，

由解的结构定理得方程的通解为

$$y = C_1 + C_2 x^2 + \frac{1}{x}.$$

**例4** 设函数 $f(u)$ 具有二阶连续导数, 而

$$z = f(e^x \sin y) \text{ 满足方程 } \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = e^{2x} z, \text{ 求 } f(u).$$

**解**  $z_x = f'(e^x \sin y) e^x \sin y$

$$z_y = f'(e^x \sin y) e^x \cos y$$

$$z_{xx} = e^x \sin y f'(e^x \sin y) + e^{2x} \sin^2 y f''(e^x \sin y)$$

$$z_{yy} = -e^x \sin y f'(e^x \sin y) + e^{2x} \cos^2 y f''(e^x \sin y)$$

$$\therefore \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = e^{2x} f''(e^x \sin y) = e^{2x} z = e^{2x} f,$$

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$$\text{即, } f''(u) = f(u) \Rightarrow f(u) = c_1 e^u + c_2 e^{-u}$$

# 课堂练习

$$1. y'' - y' = 2(1 - x) \quad y|_{x=0} = 1, y'|_{x=0} = 1$$

$$2. y'' + y = \cos x \cos 2x$$

$$3. y'' - 2y' - 8y = 5e^x \sin 2x$$

$$4. xy'' + 2y' = 12 \ln x$$

5 已知  $u(x, y, z) = u(\sqrt{x^2 + y^2 + z^2})$  有连续的二阶导数, 且满足  $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$ , 求  $u(x, y, z)$

6 设  $f(x)$  连续满足  $f(x) = \sin x + \int_0^x (\int_0^t f(u) du) dt$   
求  $f(u)$

$$1. y'' - y' = 2(1 - x) \quad y|_{x=0} = 1, y'|_{x=0} = 1$$

解法1

$$\text{令 } y' = p \quad \frac{dp}{dx} - p = 2(1 - x)$$

$$p = e^{\int dx} (c_1 + \int (2 - 2x)e^{-x} dx) = e^x (c_1 + 2xe^{-x})$$

$$\text{由 } y'|_{x=0} = 1 \text{ 代入得, } c_1 = 1, \therefore p = e^x (1 + 2xe^{-x})$$

$$\text{通解: } y = e^x + x^2 + c_2$$

$$\text{由 } y|_{x=0} = 1 \text{ 代入得, } c_2 = 0, \text{ 特解 } y = e^x + x^2$$

## 解法2

$$r^2 - r = 0 \quad r = 0, r = 1 \quad Y = c_1 + c_2 e^x$$

$$\text{令 } y^* = x(b_0 + b_1 x) = b_0 x + b_1 x^2$$

$$y^{*'} = b_0 + 2b_1 x \quad y^{*''} = 2b_1$$

$$2b_1 - b_0 - 2b_1 x = 2 - 2x \Rightarrow b_1 = 1, b_0 = 0$$

$$y^* = x(0 + x)$$

$$\therefore \text{通解 } y = c_1 + c_2 e^x + x^2 \quad c_1 + c_2 = 1$$

$$y' = c_2 e^x + 2x, \text{ 代入初始条件得, } c_2 = 1, c_1 = 0$$

$$\therefore y = e^x + x^2$$



## 2. $y'' + y = \cos x \cos 2x$

解  $r^2 + 1 = 0 \quad r = \pm i \quad Y = c_1 \cos x + c_2 \sin x$

$$f(x) = \cos x \cos 2x = \frac{1}{2} [\cos 3x + \cos x]$$

$$= \frac{1}{2} \cos 3x + \frac{1}{2} \cos x$$

对于  $f_1(x) = \frac{1}{2} \cos 3x$

令  $y_1^* = (a \cos 3x + b \sin 3x)$

$$y_1^{*'} = (-3a \sin 3x + 3b \cos 3x) \quad y_1^{*''} = (-9a \cos 3x - 9b \sin 3x)$$

$$-9a \cos 3x - 9b \sin 3x + a \cos 3x + b \sin x = \frac{1}{2} \cos 3x$$

$$b = 0, a = -\frac{1}{16} \quad \therefore y_1^* = -\frac{1}{16} \cos 3x$$

对于  $f_2(x) = \frac{1}{2} \cos x$ , 令  $y_2^* = x(a \cos x + b \sin x)$

$$y_2^{*'} = a \cos x + b \sin x + x(-a \sin x + b \cos x)$$

$$y_2^{*''} = -a \sin x + b \cos x + (-a \sin x) + b \cos x + x(-a \cos - b \sin x)$$

$$= 2(b \cos x - a \sin x) - x(a \cos + b \sin x)$$

$$+ 2(b \cos x - a \sin x) - x(a \cos x + b \sin x) = \frac{1}{2} \cos x$$

$$\begin{cases} 2b = \frac{1}{2} \\ a = 0 \end{cases} \Rightarrow a = 0, b = \frac{1}{4} \quad y_2^* = x\left(\frac{1}{4} \sin x\right)$$

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$$\therefore y^* = -\frac{1}{16} \cos 3x + \frac{1}{4} x \sin x$$

$$y = Y + y^* = c_1 \cos x + c_2 \sin x - \frac{1}{16} \cos 3x + \frac{1}{4} x \sin x$$

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$$3. y'' - 2y' - 8y = 5e^x \sin 2x$$

解  $r^2 - 2r - 8 = 0 \quad r = -2, r = 4$

$$Y = c_1 e^{-2x} + c_2 e^{4x}$$

$$\text{令 } y^* = e^x (a \cos 2x + b \sin 2x)$$

$$y^{*'} = (a \cos 2x + b \sin 2x - 2a \sin 2x + 2b \cos 2x) e^x$$

$$y^{*''} = e^x (a \cos 2x + b \sin 2x - 2a \sin 2x + 2b \cos 2x - 2a \sin 2x + 2b \cos 2x - 4a \cos 2x - 4b \sin 2x)$$

$$= e^x [(4b - 3a) \cos 2x + (-4a - 3b) \sin 2x]$$

$$e^x [(4b - 3a) \cos 2x + (-4a - 3b) \sin 2x] - 2e^x [(a + 2b) \cos 2x + (b - 2a) \sin 2x] - 8e^x (a \cos 2x + b \sin 2x) = 5e^x \sin x$$

$$\left. \begin{aligned} 4b - 3a - 2(a + 2b) - 8a &= 0 \\ -4a - 3b - 2(b - 2a) - 8b &= 5 \end{aligned} \right\} \Rightarrow \begin{cases} -13a = 0 & a = 0 \\ -13b = 5 & b = -\frac{5}{13} \end{cases}$$

$$\therefore y^* = e^x \left( -\frac{5}{13} \sin 2x \right), y = c_1 e^{-2x} + c_2 e^{4x} - \frac{5}{13} \sin 2x e^x$$

## 4. $xy'' + 2y' = 12 \ln x$

**解法1** 令  $P = y'$   $P' + \frac{2}{x}P = 12 \frac{\ln x}{x}$

$$P = e^{\int -\frac{2}{x} dx} (c_1 + \int 12 \frac{\ln x}{x} e^{\int \frac{2}{x} dx})$$

$$= \frac{1}{x^2} (c_1 + \int 12x \ln x dx)$$

$$= \frac{1}{x^2} (c_1 + 6x^2 \ln x - \int 6x dx)$$

$$= \frac{1}{x^2} (c_1 + 6x^2 \ln x - 3x^2) = \frac{c_1}{x^2} + 6 \ln x - 3$$

$$y' = \frac{c_1}{x^2} + 6 \ln x - 3$$

$$y = -\frac{c_1}{x} - 3x + 6x \ln x - \int 6dx$$

$$= -\frac{c_1}{x} - 3x + 6x \ln x - 6x + c_2$$

**解法2**  $x^2 y'' + 2xy' = 12x \ln x$

$$\text{令 } x = e^t \quad t = \ln x \quad y'' = \frac{1}{x} D(D-1)y \quad y' = \frac{1}{x} D$$

$$D(D-1)y + 2Dy = 12e^t t$$

$$D(D+1)y = 12te^t \quad \text{即} \frac{d^2 y}{dt^2} + \frac{dy}{dt} = 12te^t$$

$$r^2 + r = 0 \quad r = 0, r = -1 \quad Y = c_1 + c_2 e^{-t}$$



$$\text{令 } y^* = (b_0 + b_1 t)e^t$$

$$y^{*'} = e^t (b_0 + b_1 x + b_1)$$

$$y^{*''} = e^t (b_0 + b_1 + b_1 x + b_1)$$

$$e^t (b_0 + 2b_1 + b_1 t) + e^t (b_0 + b_1 + b_1 t) = 12te^t$$

$$\begin{cases} 2b_0 + 3b_1 = 0 \\ 2b_1 = 12 \quad b_1 = 6 \quad b_0 = -9 \end{cases}$$

$$\therefore y^* = (-9 + 6t)e^t$$

$$y = c_1 + c_2 \frac{1}{x} - 9x + 6e^x \ln x$$

5 已知  $u(x, y, z) = u(\sqrt{x^2 + y^2 + z^2})$  有连续的二阶导数, 且满足  $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$ , 求  $u(x, y, z)$

解

$$\frac{\partial u}{\partial x} = u'(r) \frac{2x}{2\sqrt{x^2 + y^2 + z^2}} = \frac{xu'(r)}{r}$$

$$\frac{\partial u}{\partial y} = \frac{yu'(r)}{r} \quad \frac{\partial u}{\partial z} = \frac{zu'(r)}{r}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{u'(r)}{r} + x \frac{u''(r)r - u'}{r^2} \cdot \frac{x}{r} = \frac{u'}{r} + \frac{x^2}{r^3} (ru'' - u')$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{3u'}{r} + \frac{ru'' - u'}{r^3} r^2 = 0$$

$$u' = p \quad \frac{2}{r}p + p' = 0 \quad \frac{dp}{dr} = -\frac{2}{r}p \quad \frac{dp}{p} = -\frac{2}{r}dr$$

$$\ln p = -2\ln r + \ln c_1 \quad p = \frac{c_1}{r^2}, \quad \frac{du}{dr} = \frac{c_1}{r^2} \quad u = -\frac{c_1}{r} + c_2$$

$$\text{即 } u(x, y, z) = \frac{c_1}{\sqrt{x^2 + y^2 + z^2}} + c_2$$

6 设 $f(x)$ 连续满足 $f(x) = \sin x + \int_0^x (\int_0^t f(u) du) dt$   
求 $f(u)$

解  $f(0) = 0$        $f'(x) = \cos x + \int_0^x f(u) du$

$$f'(0) = 1 \quad f''(x) = -\sin x + f(x)$$

$$i, e \quad y'' = -\sin x + y \Rightarrow y'' - y = -\sin x$$

$$r^2 - 1 = 0 \quad r = 1, r = -1 \quad Y = c_1 e^x + c_2 e^{-x}$$

$$\text{令 } y^* = a \cos x + b \sin x$$

$$y^{*'} = -a \sin x + b \cos x \quad y^{*''} = -a \cos x - b \sin x$$

$$\text{代入得, } -a \cos x - b \sin x - (a \cos x + b \sin x) = -\sin x,$$

$$-2a \cos x - 2b \sin x = -\sin x$$

$$a = 0, b = \frac{1}{2} \quad y^* = -\frac{1}{2} \sin x$$

$$\therefore y = c_1 e^x + c_2 e^{-x} + \frac{1}{2} \sin x$$

$$c_1 + c_2 = 0, c_1 = -c_2$$

$$y' = c_1 e^x - c_2 e^{-x} + \frac{1}{2} \cos x \quad 1 = c_1 - c_2 + \frac{1}{2} \quad 2c_1 = \frac{1}{2}, c_1 = \frac{1}{4}$$

$$y = \frac{1}{4} (e^x - e^{-x}) + \frac{1}{2} \sin x$$

**例7** 设函数 $f(u)$ 具有二阶连续导数, 而

$$z = f(e^x \sin y) \text{ 满足方程 } \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = e^{2x} z, \text{ 求 } f(u).$$

**解**  $z_x = f'(e^x \sin y) e^x \sin y$

$$z_y = f'(e^x \sin y) e^x \cos y$$

$$z_{xx} = e^x \sin y f'(e^x \sin y) + e^{2x} \sin^2 y f''(e^x \sin y)$$

$$z_{yy} = -e^x \sin y f'(e^x \sin y) + e^{2x} \cos^2 y f''(e^x \sin y)$$

$$\therefore \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = e^{2x} \cancel{f''}(e^x \sin y) = e^{2x} z = e^{2x} \cancel{f},$$

---

$$\text{即, } f''(u) = f(u) \Rightarrow f(u) = C_1 e^u + C_2 e^{-u}$$

**例8** (1) 试确定以 $y = \sin 2x$ 为特解的二阶常系数线性齐次微分方程。

(2) 试确定以 $y_1 = e^x, y_2 = xe^x, y_3 = 2\cos 2x, y_4 = \sin 2x$ 为特解的四阶常系数线性齐次微分方程。

分析：这是求解常系数 线性齐次微分方程的反问题。求解这类问题的方法是：

*i*)由特解确定出特征根；

*ii*)由特征根导出特征方程 ；

*iii*)由特征方程导出常系数 线性齐次方程。

**解** (1) *i*)特征根为 $r_1 = 2i \Rightarrow r_2 = -2i$

*ii*)特征方程为 $(r - 2i)(r + 2i) = 0$ ,

$$\text{即 } r^2 + 4 = 0$$

*iii*)所求齐次线性方程:  $y'' + 4y = 0$

(2) *i*)由特解  $y_1 = e^x, y_2 = xe^x$  知  $r_{1,2} = 1$  (二重根)

由特解  $y_3 = 2 \cos 2x, y_4 = \sin 2x$  知,  $r_{3,4} = \pm 2i$

$$\text{ii}) (r - 1)^2 (r - 2i)(r + 2i) = 0$$

$$\text{即 } r^4 - 2r^3 + 5r^2 - 8r + 4 = 0$$

*iii*)所求齐次线性方程:

---

$$y^{(4)} - 2y''' + 5y'' - 8y' + 4y = 0$$



**Ex1.** 已知 $x = \sin t$ 是二阶常系数线性齐次微分方程的一个特解,求此微分方程.

提示: *i*)由题意知特征根为 $r_1 = i \Rightarrow r_2 = -i$

*ii*)特征方程为 $(r - i)(r + i) = 0$ , 即 $r^2 + 1 = 0$

*iii*)所求齐次线性方程:  $y'' + 4y = 0$

**Ex2.** 求以 $\sin 2t, te^t$ 为两个特解的四阶常系数线性齐次微分方程.

提示: *i)*由题意知特征根为 $r_1 = r_2 = 1$ (二重),

$$r_3 = 2i \Rightarrow r_4 = -2i$$

*ii)*特征方程为 $(r - 2i)(r + 2i)(r - 1)^2 = 0$ ,

即 $(r^2 + 4)(r - 1)^2 = 0$ , 即 $r^4 - 2r^3 + 5r^2 - 8r + 4 = 0$

*iii)*所求齐次线性方程:

$$y^{(4)} - 2y^{(3)} + 5y'' - 8y' + 4y = 0$$

## 例9

设  $f(x)$  具有二阶连续导数且  $f(0) = 0, f'(0) = 1$   
又已知曲线积分

$$\int_L (xe^{2x} - 6f(x)) \sin y dx - (5f(x) - f'(x)) \cos y dy$$

与路径无关，求  $f(x)$

解：由题意有  $\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$

即有  $f''(x) - 5f'(x) + 6f(x) = xe^{2x}$  \*

$Y = C_1e^{2x} + C_2e^{3x}$ , 设  $y^* = x(ax + b)e^{2x}$

将  $y^*, y^{*''}$  代入方程\*  $\Rightarrow a = -\frac{1}{2}, b = -1$

$\therefore f(x) = C_1e^{2x} + C_2e^{3x} - \frac{x}{2}(x + 2)e^{2x}, f'(x) = \dots$

再有  $f(0) = 0, f'(0) = 1 \Rightarrow C_1 = -2, C_2 = 2$

故  $\therefore f(x) = -2e^{2x} + 2e^{3x} - \frac{x}{2}(x + 2)e^{2x}$

**Ex.** 16. 设  $f(x)$  具有二阶连续导数, 且  $f(0)=0, f'(0)=1$ , 又已知曲线积分

$\int_L (xe^{2x} - 6f(x)) \sin y dx - (5f(x) - f'(x)) \cos y dy$  与路径无关, 求  $f(x)$ .

解: 设  $P = (xe^{2x} - 6f(x)) \sin y, Q = -(5f(x) - f'(x)) \cos y$ , 由曲线积分与路径无关知  $\frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y}$ , 即  $f''(x) - 5f'(x) + 6f(x) = xe^{2x}$ . 其所对应的特征方程为  $\lambda^2 - 5\lambda + 6 = 0$ , 解得  $\lambda = 2, \lambda = 3$ . 又由  $u=2, P(x) = x$  知待求特解形式为  $f^*(x) = x(A_0 + A_1 x)e^{2x}$ , 将  $f^*(x)$  代入方程  $f''(x) - 5f'(x) + 6f(x) = xe^{2x}$ , 比较系数可解得  $A_0 = -1, A_1 = -\frac{1}{2}$ , 故方程  $f''(x) - 5f'(x) + 6f(x) = xe^{2x}$  的通解为  $f(x) = C_1 e^{2x} + C_2 e^{3x} - \frac{x}{2}(x+2)e^{2x}$ , 又由  $f(0)=0, f'(0)=1$  可得  $\begin{cases} C_1 + C_2 = 0 \\ 2C_1 + 3C_2 = 2 \end{cases}$ , 故  $\begin{cases} C_1 = -2 \\ C_2 = 2 \end{cases}$ , 从而所求  $f(x) = -2e^{2x} + 2e^{3x} - \frac{x}{2}(x+2)e^{2x}$ .

17. 设函数  $f(u)$  具有二阶连续导数, 函数  $z = f(e^x \sin y)$  满足方程  $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = (z+1)e^{2x}$ , 若

$f(0)=0, f'(0)=0$ , 求函数  $f(u)$  的表达式.

解: 设  $u = e^x \sin y$ , 则  $z = f(e^x \sin y) = f(u)$ , 故  $\frac{\partial z}{\partial x} = f'(u)e^x \sin y, \frac{\partial^2 z}{\partial x^2} = f''(u)e^{2x} \sin^2 y + f'(u)e^x \sin y$ ,  $\frac{\partial z}{\partial y} = f'(u)e^x \cos y, \frac{\partial^2 z}{\partial y^2} = f''(u)e^{2x} \cos^2 y - f'(u)e^x \sin y$ , 从而  $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = f''(u)e^{2x} = (f(u)+1)e^{2x}$ , 即为  $f''(u) - f(u) = 1$ , 其所对应的特征方程为  $\lambda^2 - 1 = 0$ , 解得  $\lambda_1 = 1, \lambda_2 = -1$ , 又由  $u=0, P(u) = 1$  知待求特解形式为  $f^*(u) = A$ , 将  $f^*(u)$  代入方程  $f''(u) - f(u) = 1$  可知  $A = -1$ . 从而方程  $f''(u) - f(u) = 1$  的通解为  $f(u) = C_1 e^u + C_2 e^{-u} - 1$ , 又由  $f(0)=0, f'(0)=0$  可解得  $C_1 = C_2 = \frac{1}{2}$ , 故所求  $f(u) = \frac{1}{2}(e^u + e^{-u}) - 1$ .

**例10** 求微分方程  $y'' + y = 4x \sin x$  的通解

**解** 
$$f(x) = e^{\alpha x} [P_l(x) \cos \beta x + P_n(x) \sin \beta x]$$

$$\alpha = 0, \beta = 1, P_n(x) = 4x, P_l = 0$$

$$\text{特征方程 : } r^2 + 1 = 0, \quad \text{特征根: } r = \pm i$$

$$\text{令 } y^* = x[(a + bx) \cos x + (c + dx) \sin x]$$

$$y^{*'} = [a + (2b + c)x + dx^2] \cos x + [c + (2d - a)x - bx^2] \sin x$$

$$y^{*''} = \cos x[(2b + 2c) + (4d - a)x - bx^2]$$

$$+ \sin x[(2d - 2a) - (4b + c)x - dx^2] \text{ 代入原方程}$$

$$\underline{[2(b + c) + 4dx] \cos x + \sin x[2(d - a) - 4bx] = 4x \sin x}$$

$$\begin{cases} 2b + 2c + 4dx = 0 \\ (2d - 2a) + (-4b)x = 4x \end{cases} \quad \begin{cases} d = 0 & b + c = 0 \\ a = 0 & b = -1 & c = 1 \end{cases}$$

代入  $y^* = x[(a + bx)\cos x + (c + dx)\sin x]$

$$\therefore y^* = x(-x)\cos x + x\sin x = -x^2 \cos x + x\sin x$$

通解:  $y = c_1 \cos x + c_2 \sin x + x\sin x - x^2 \cos x$

**例11** 求微分方程  $y'' + y = \underbrace{4x \sin x}_{f_2(x)} + \underbrace{x^2 + 1}_{f_1(x)}$  的通解

**解** 由例 1,  $y_1^* = x^2 - 1$

是  $y'' + y = x^2 + 1 = f_1(x)$  的特解

由例 6,  $y_2^* = x \sin x - x^2 \cos x$

是  $y'' + y = 4x \sin x = f_2(x)$  的特解

故  $y_1^* + y_2^* = x^2 - 1 + x \sin x - x^2 \cos x$

是  $y'' + y = 4x \sin x + x^2 + 1$  的特解

$\therefore$  通解:  $y = c_1 \cos x + c_2 \sin x + x^2 - 1 + x \sin x - x^2 \cos x$



**Ex.** 如何求方程  $y'' - y' = x + e^{2x} + \sin x$   
的一个特解  $y^*$ ?

答: 分别求出  $y'' - y' = x$ ;  $y'' - y' = e^{2x}$ ;

$y'' - y' = \sin x$  的特解  $y_1^*$ 、 $y_2^*$  和  $y_3^*$ , 则

$y^* = y_1^* + y_2^* + y_3^*$  是  $y'' - y' = x + e^{2x} + \sin x$  的特解.

## 例12 解方程 $y'' + y = \cos x \cos 2x$ .

**解**  $r^2 + 1 = 0 \quad r = \pm i \quad Y = C_1 \cos x + C_2 \sin x$

$$f(x) = \cos x \cos 2x = \frac{1}{2} [\cos 3x + \cos x]$$

$$= \frac{1}{2} \cos 3x + \frac{1}{2} \cos x$$

对于  $f_1(x) = \frac{1}{2} \cos 3x$

令  $y_1^* = (a \cos 3x + b \sin 3x)$

$$y_1^{*'} = (-3a \sin 3x + 3b \cos 3x) \quad y_1^{*''} = (-9a \cos 3x - 9b \sin 3x)$$

$$-9a \cos 3x - 9b \sin 3x + a \cos 3x + b \sin x = \frac{1}{2} \cos 3x$$

$$b = 0, a = -\frac{1}{16} \quad \therefore y_1^* = -\frac{1}{16} \cos 3x$$

对于  $f_2(x) = \frac{1}{2} \cos x$ , 令  $y_2^* = x(a \cos x + b \sin x)$

$$y_2^{*'} = a \cos x + b \sin x + x(-a \sin x + b \cos x)$$

$$y_2^{*''} = -a \sin x + b \cos x + (-a \sin x) + b \cos x + x(-a \cos - b \sin x)$$

$$= 2(b \cos x - a \sin x) - x(a \cos + b \sin x)$$

$$+ 2(b \cos x - a \sin x) - x(a \cos x + b \sin x) = \frac{1}{2} \cos x$$

$$\begin{cases} 2b = \frac{1}{2} \\ a = 0 \end{cases} \Rightarrow a = 0, b = \frac{1}{4} \quad y_2^* = x\left(\frac{1}{4} \sin x\right)$$

$$\therefore y^* = -\frac{1}{16} \cos 3x + \frac{1}{4} x \sin x$$

$$y = Y + y^* = C_1 \cos x + C_2 \sin x - \frac{1}{16} \cos 3x + \frac{1}{4} x \sin x$$

**Ex1.** 求微分方程  $y''+4y = 2\cos^2 x$  的通解

例13 设 $f(x)$ 连续满足 $f(x) = \sin x + \int_0^x (\int_0^t f(u) du) dt$   
求 $f(x)$

解  $f(0) = 0$        $f'(x) = \cos x + \int_0^x f(u) du$

$$f'(0) = 1 \quad f''(x) = -\sin x + f(x)$$

$$i, e \quad y'' = -\sin x + y \Rightarrow y'' - y = -\sin x$$

$$r^2 - 1 = 0 \quad r = 1, r = -1 \quad Y = c_1 e^x + c_2 e^{-x}$$

$$\text{令 } y^* = a \cos x + b \sin x$$

$$y^{*'} = -a \sin x + b \cos x \quad y^{*''} = -a \cos x - b \sin x$$

$$\text{代入得, } -a \cos x - b \sin x - (a \cos x + b \sin x) = -\sin x,$$

$$-2a \cos x - 2b \sin x = -\sin x$$

$$a = 0, b = \frac{1}{2} \quad y^* = -\frac{1}{2} \sin x$$

$$\therefore y = C_1 e^x + C_2 e^{-x} + \frac{1}{2} \sin x, \text{ 则}$$

$$C_1 + C_2 = 0, C_1 = -C_2$$

$$y' = C_1 e^x - C_2 e^{-x} + \frac{1}{2} \cos x, \text{ 则 } 1 = C_1 - C_2 + \frac{1}{2}, \quad 2C_1 = \frac{1}{2}, C_1 = \frac{1}{4}$$

$$y = f(x) = \frac{1}{4}(e^x - e^{-x}) + \frac{1}{2} \sin x$$

**Ex2.** 设  $f(x) = \sin x - \int_0^x (x-t)f(t)dt$ , 其中  $f(x)$  为连续函数, 求  $f(x)$ .



例 14 已知  $y_1 = 3$ ,  $y_2 = 3 + x^2$ ,  $y_3 = 3 + x^2 + e^x$  都是微分方程  $(x^2 - 2x)y'' - (x^2 - 2)y' + (2x - 2)y = 6(x - 1)$  的解, 求此方程所对应齐次方程的通解.

**解**  $\because y_1, y_2, y_3$  都是微分方程的解,

$$\therefore y_3 - y_2 = e^x, \quad y_2 - y_1 = x^2,$$

是对应齐次方程的解,

$$\therefore \frac{y_3 - y_2}{y_2 - y_1} = \frac{e^x}{x^2} \neq \text{常数}$$

$$\begin{aligned} \therefore \text{所求通解为 } y &= C_1(y_3 - y_2) + C_2(y_2 - y_1) \\ &= C_1 e^x + C_2 x^2. \end{aligned}$$

## 复习

例 试确定以 $y = \sin 2x$ 为特解的二阶常系数线性齐次微分方程。

分析：这是求解常系数 线性齐次微分方程的反问题。求解这类问题的方法是：

*i)*由特解确定出特征根；

*ii)*由特征根导出特征方程 ；

*iii)*由特征方程导出常系数 线性齐次方程。

解 *i*)特征根为 $r_1 = 2i \Rightarrow r_2 = -2i$

*ii*)特征方程为 $(r - 2i)(r + 2i) = 0$ ,

$$\text{即 } r^2 + 4 = 0$$

*iii*)所求齐次线性方程:  $y'' + 4y = 0$

### Ex3.

已知  $y_1 = xe^x + e^{2x}$ ,  $y_2 = xe^x + e^{-x}$ ,  $y_3 = xe^x + e^{2x} - e^{-x}$  是二阶常系数线性非齐次微分方程的三个特解, 求此微分方程及通解.

**例15** 已知二阶常微分方程  $y'' + ay' + by = ce^x$  有特解  $y = e^{-x}(1 + xe^{2x})$ , 求微分方程的通解.

**解:** 将特解代入方程得恒等式

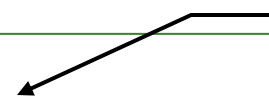
$$(1-a+b)e^{-x} + (2+a)e^x + (1+a+b)xe^x = ce^x$$

比较系数得 
$$\begin{cases} 1-a+b=0 \\ 2+a=c \\ 1+a+b=0 \end{cases} \longrightarrow \begin{cases} a=0 \\ b=-1 \\ c=2 \end{cases}$$

故原方程为  $y'' - y = 2e^x$

对应齐次方程通解:  $Y = C_1 e^x + C_2 e^{-x}$

原方程通解为  $y = C_1 e^x + C_2 e^{-x} + xe^x$

$$y = e^{-x} + xe^x$$


**Ex4.** 指出下列方程的特解形式是什么？

$$y'' - y' = x \qquad y^* = x(a + bx)$$

$$y'' - y' = e^{2x} \qquad y^* = be^{2x}$$

$$y'' + y' = \sin x \qquad y^* = (a \cos + b \sin x)$$

$$y'' + y' = x^2 \qquad y^* = x(b_0 + b_1x + b_2x^2)$$

例16 设函数  $f(t)$  在  $[0, +\infty)$  上连续, 且满足方程

$$f(t) = e^{4\pi t^2} + \iint_{x^2 + y^2 \leq 4t^2} f\left(\frac{1}{2}\sqrt{x^2 + y^2}\right) dx dy,$$

求  $f(t)$ .

提示: 参见下文例16-1

例 16-1 求  $F'(t)$ , 设

$$(2) F(t) = \iiint_{x^2+y^2+z^2 \leq t^2} f(x^2 + y^2 + z^2) dV, \quad \text{其中 } f(u) \text{ 为可微函数};$$

解

$$\begin{aligned} F(t) &= \iiint_{x^2+y^2+z^2 \leq t^2} f(x^2 + y^2 + z^2) dV \\ &= \int_0^{2\pi} d\theta \int_0^\pi \sin \varphi d\varphi \int_0^t f(r^2) r^2 dr \\ &= 4\pi \int_0^t f(r^2) r^2 dr \end{aligned}$$

$$F'(t) = 4\pi t^2 f(t^2)$$